

Optical Instrumentation Lab

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

5089B

Credits

4

Periods

5/week

Course Details

Title: Optical Instrumentation Lab

Department: Common

Revision: REV_Optical Instrumentation Lab

Pattern: Theory / Practical

Complies with official SITTTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 5089B — Optical Instrumentation Lab**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Malayalam support: ു ഐഐഐഐഐഐ ഐഐഐഐ ഐഐഐഐ English technical terms- ഐഐഐഐഐഐ ഐഐഐഐഐഐ. ുഐ ഐഐഐഐഐ ഐഐഐഐഐഐ ഐഐഐഐഐഐ, formula, diagram, units, definitions ഐഐഐഐ ഐഐഐഐഐഐഐഐ ഐഐഐഐഐഐഐഐ.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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Full Answer Key

OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Optical Instrumentation Lab.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of Optical Instrumentation Lab.

Study

Detailed analysis of core principles and theoretical foundations.

Malayalam Support Note: ഓപ്റ്റിക്കൽ ഉപകരണങ്ങൾക്ക് പൊതുവായ പ്രവർത്തനം ഉണ്ടാകാൻ പരസ്പരം ആശയവിനിമയം വഴിയാണ്. ഓപ്റ്റിക്കൽ ഉപകരണങ്ങൾക്ക് പരസ്പരം ആശയവിനിമയം വഴിയാണ്.

MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: ഓപ്റ്റിക്കൽ ഉപകരണങ്ങൾക്ക് പരസ്പരം ആശയവിനിമയം വഴിയാണ്. ഓപ്റ്റിക്കൽ ഉപകരണങ്ങൾക്ക് പരസ്പരം ആശയവിനിമയം വഴിയാണ്.

MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: ഐക്യരൂപം സാങ്കേതികവിദ്യയുടെ അടിസ്ഥാനപരമായ അളവുകോലാണ്. ഐക്യരൂപം സാങ്കേതികവിദ്യയുടെ അടിസ്ഥാനപരമായ അളവുകോലാണ്.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: ഐക്യരൂപം സാങ്കേതികവിദ്യയുടെ അടിസ്ഥാനപരമായ അളവുകോലാണ്. ഐക്യരൂപം സാങ്കേതികവിദ്യയുടെ അടിസ്ഥാനപരമായ അളവുകോലാണ്.

Practice Model Question Paper

Time: 3 Hours | Max Marks: 75

Part A (2 marks each)

1. Define core concepts of Optical Instrumentation Lab.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.